Yaroslav Vorobyov
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EDUCATION

Texas Christian University – Burnett School of Medicine — Fort Worth, TX
Doctor of Medicine — July 2025 – Present

Northwestern University — Evanston, IL
Bachelor of Science in Chemistry and Neuroscience — Sep 2021 – June 2025

- Cumulative GPA: 3.95/4.0
- Dean's List, 12 quarters (2021–2025)

RESEARCH EXPERIENCE

Samuel I. Stupp Laboratory, Northwestern University — Evanston, IL
Undergraduate NMR Analyst and Chemist — Nov 2022 – June 2025

- Investigated molecular motion and structure–function relationships in peptide amphiphile supramolecular biomaterials for regenerative medicine using solution- and solid-state NMR.
- Led a sub-project using isotope-labeling strategies and multinuclear NMR to probe peptide amphiphile mechanisms; also performed peptide synthesis and SEM, AFM, and TEM characterization.
- Research contributed to 2026 publications in the Journal of the American Chemical Society and ACS Biomaterials Science & Engineering.

Clinical & Translational Research — Current
Interventional Radiology / Vascular Surgery

- Normal-Range Sodium Variability After Arthroplasty – evaluated whether normal-range sodium variability predicts skilled nursing facility discharge after hip and knee arthroplasty; coauthored a 2026 conference abstract and poster; manuscript submission in progress.
- Calcified Infrapopliteal Intervention – developed a morphology- and response-based educational framework for calcified infrapopliteal disease in chronic limb-threatening ischemia; educational exhibit accepted to ISET 2027.
- Sports Medicine Fellowship Directors – national cross-sectional study of the demographic, training, and scholarly profiles of U.S. orthopaedic sports medicine fellowship directors; manuscript under review at JBJS Open Access.
- Median Arcuate Ligament Syndrome – first-author outcomes research; study in progress.
- Aortofemoral Bypass Outcomes in Women – first-author vascular outcomes research; study in progress.

CLINICAL & SERVICE EXPERIENCE

Mercy Clinic — Fort Worth, TX
Medical Student Volunteer — Sep 2025 – Present

- Provide clinical support at a free clinic serving uninsured adults by assisting with patient histories, physical exams, and health education under physician supervision.
- Work alongside an interdisciplinary team to coordinate care and address barriers to accessing healthcare.

Superior Ambulance — Evanston, IL
Emergency Medical Technician — Jan 2023 – Jan 2025

- Provided patient transport and emergency care in a private ambulance setting while developing bedside communication, documentation, and patient-care skills across varied clinical situations.
- Completed EMT training through Loyola University Medical Center and earned National EMS certification.

New Age Elder Care — Chicago, IL
Medicaid Caregiver — Aug 2022 – June 2025

- Supported older adults with medication adherence, medical appointments, home safety, companionship, and activities of daily living.

Ascension Saint Francis — Evanston, IL
Floor Secretary Assistant Volunteer — June 2022 – Aug 2024

- Assisted nursing staff with specimen transport, vital-sign collection, PPE restocking, patient support, and organizational tasks.

Simply From the Heart — Chicago, IL
Glam Squad Volunteer — Jan 2024 – June 2025

- Supported long-term care patients through pampering and social activities at hospitals and community pop-up events.

Academy of Music and Arts for Special Education
Harp Teacher — Feb 2023 – Jan 2025

- Provided individualized music instruction to children with disabilities while fostering confidence, creativity, and self-expression.

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LEADERSHIP & PROFESSIONAL INVOLVEMENT

WCVIS Texas
Education Chair June 2026 – Present
- Launched and lead VIR Decoded, the chapter's education series introducing trainees to the IR consult, procedural planning, and life in the VIR suite.
- Develop chapter content and promote the regional symposium to expand access to vascular interventional radiology education.

TNVIRJC
Digital Media Co-Chair Aug 2026 – Present
- Lead digital communications and support virtual journal club programming, online logistics, and content connecting trainees with current interventional radiology literature.

Surgery SIG, TCU Burnett School of Medicine Fort Worth, TX
Experiential Learning Lead May 2026 – Present
- Coordinate specialty talks, suturing sessions, Step 1 review, and cadaver-based activities that connect preclinical learning with surgical skills and specialty exposure.

Business in Medicine SIG, TCU Burnett School of Medicine Fort Worth, TX
Vice President (Member since Oct 2025) June 2026 – Present
- Coordinate programming at the intersection of business and medicine, including speaker events and educational initiatives focused on healthcare management, finance, and the business aspects of medical practice.

HEENT SIG, TCU Burnett School of Medicine Fort Worth, TX
Events Chair Oct 2025 – July 2026
- Organized specialty-focused physician speaker events from initial concept through execution, including outreach, scheduling, promotion, and event logistics.

WCVIS
Advertising Chair, WCVIS 2027 Symposium (Member since May 2026) Feb 2027
- Lead event advertising and social media outreach, designing on-brand flyers, the symposium program, and speaker spotlights.
- Help recruit trainee cases for the Angio Club competition and provide live coverage during the symposium.

Academy 4 Fort Worth, TX
Mentor Sep 2025 – May 2026
- Mentored fourth-grade students through monthly leadership lessons focused on communication, teamwork, empathy, character development, civic engagement, and confidence.

Texas Hip and Knee Foundation Fort Worth, TX
Member Aug 2025 – Present
- Participate in educational sessions, research presentations, panels, and professional discussions focused on joint reconstruction, clinical decision-making, healthcare systems, and academic orthopedics.

ADDITIONAL LEADERSHIP & ACTIVITIES

Northwestern University Marching Band / Marching Band Staff Evanston, IL
Field and Properties Manager Aug 2021 – Aug 2023
- Managed band equipment, uniforms, and instruments and evaluated field setup and strike workflows to improve efficiency.

Northwestern Concert, Purple, and Basketball Bands Evanston, IL
Clarinetist Aug 2021 – June 2025
- Performed as a clarinetist across concert and athletic ensemble settings.

Peer Study Group Program, Northwestern University Evanston, IL
Facilitator / Senior Facilitator Sep 2022 – June 2024
- Facilitated active learning for General Chemistry and Biology and mentored fellow facilitators through session preparation and reflection.

Additional organization involvement: Founded a United Students for Veterans' Health chapter at Northwestern (Recruitment Chair and Social Media Chair); Phi Delta Epsilon; NUAviation Club (Communications Chair); Biology Lab and Neuroscience teaching assistant; Catholic Student Association (Lector).

PUBLICATIONS & PRESENTATIONS

1. Pavlović R. Z.; **Vorobyov Y.;** et al. *Probing Supramolecular Motion with Nuclear Magnetic Resonance in Bioactive Scaffolds that Promote Neural Regeneration in the Central Nervous System.* Journal of the American Chemical Society. 2026. **Equal-contributing author.** https://doi.org/10.1021/jacs.6c01684
2. Metlushko A.; Sather N. A.; Fyrner T.; Takata N.; **Vorobyov Y.;** et al. *Design of Neuronal Supramolecular Scaffolds Integrating Cell Signaling and Electrical Conductivity.* ACS Biomaterials Science & Engineering. 2026. https://doi.org/10.1021/acsbiomaterials.5c01720

3. Badr O.; Shakthy E.; **Vorobyov Y.;** Shau D. *Normal-Range Sodium Variability Is Not Associated With Skilled Nursing Facility Discharge After Total Hip and Knee Arthroplasty in Elderly Patients.* THK Conference. April 2026. Conference abstract and poster. https://yaroslavvorobyov.com/arthroplasty-poster.pdf
4. Bannikov S.; **Vorobyov Y.;** Ming D.; Khatri S.; Vatakencherry G. *A Morphology- and Response-Based Framework for Calcified Infrapopliteal Intervention in CLTI.* ISET 2027. Educational exhibit, accepted 2026. https://yaroslavvorobyov.com/iset-abstract.pdf
5. Berghult C.; **Vorobyov Y.;** Ani K.; Mellberg J.; Shau D. *The Leadership Phenotype of Orthopaedic Sports Medicine Fellowship Directors: A National Cross-Sectional Study.* JBJS Open Access. Manuscript under review, submitted 2026.
6. Pavlović R. Z.; **Vorobyov Y.;** et al. *Nuclear Magnetic Resonance Insights into Supramolecular Structuring and Motion in Biomaterials for Neuronal and Chondrogenic Regeneration.* Experimental Nuclear Magnetic Resonance Conference. Poster presentation, April 2025. https://yaroslavvorobyov.com/enc-abstract.pdf
7. Pavlovic R.; Syrgiannis Z.; Egner S.; Metlushko A.; **Vorobyov Y.;** Singer E.; Hansen A. L.; Yuan C.; Wang S.; Waudby C.; Synatschke C.; Graf R.; Stupp S. I. *Deciphering Dynamics of Supramolecular Polymers with Multiphase and Multinuclear NMR Spectroscopy.* ACS Fall 2023, San Francisco, CA. Conference abstract and poster. https://yaroslavvorobyov.com/acs-poster.pdf

PROFESSIONAL MEMBERSHIPS

Society of Interventional Radiology (SIR); Association of Academic Radiology (AAR); American College of Radiology (ACR); American College of Surgeons (ACS).

HONORS, CERTIFICATIONS & SKILLS

Awards: Dean's List, 12 quarters (2021–2025); Summer Undergraduate Research Grant (2023, 2024). Clatterbuck Family Scholarship — Burnett School of Medicine at TCU, merit-based competitive scholarship, 2025–2026. TCU SOM Annual Scholarship — Burnett School of Medicine at TCU, institutional scholarship, 2025–Present

Certifications: NREMT EMT-B Certification; Illinois State EMT Licensure; BLS CPR Certification.

Languages: English and Russian (fluent); Polish (basic).

Laboratory & analytical: Powder analysis; purification techniques; optical microscopy; mass spectrometry (including MALDI-TOF MS); NMR spectroscopy (solution and solid-state); IR and FTIR spectroscopy; Raman spectroscopy; UV/Vis spectroscopy; fluorescence spectroscopy; X-ray crystallography; electrochemistry; atomic spectroscopy; gel permeation chromatography; differential scanning calorimetry; SEM; AFM.

Synthesis & methods: Organic and inorganic synthesis; multi-step synthesis; reaction mechanism analysis; kinetic analysis; microscale chemistry; Schlenk line techniques; preparation of cells for analysis.

Software & professional skills: MestreNova; TopSpin; R; Racket; Microsoft Office; Trello; Adobe Illustrator; technical writing; literature review; research writing; drafting; analytical skills; data management; critical thinking.

Interests: Aviation and flight simulation; photography; automotive and mechanical projects; sauna/banya society and culture; weightlifting; clarinet.